

Class: XII

F.M: 75

Time: 3hrs.

Subject: Mathematics [0081]

P.M: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - A [MCQ] [11×1=11]

1. In how many ways can 4 different books be arranged on a shelf?
a) 4 b) 8 c) 24 d) 16
2. During the process of solving system of the equation in x, y, z if we get $0 \cdot z = 0$, then the system is
a) inconsistent and has no solution
b) consistent and has unique solution
c) inconsistent but infinite solution
d) consistent and has infinitely solution
3. The number of terms in the expansion $(a+x)^{n+1}$, n is a positive integer, is:
a) n b) $n+1$ c) $n+2$ d) ∞
4. If ω is cube root of unity, then the value of $(1-\omega+\omega^2)^3$ is:
a) 0 b) 1 c) ω d) -8
5. The value of $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec 3x}{\sec x}$ is.....
a) $\frac{1}{3}$ b) $-\frac{1}{3}$ c) 0 d) none
6. If $C(n, 2) = C(n, 3)$ then the value of $C(n, 4)$ is ...
a) 1 b) 5 c) 3 d) 4
7. Which is the derivative of $\tanh^{-1}x$?
a) $\frac{1}{\sqrt{1+x^2}}$ b) $\frac{1}{1+x^2}$ c) $\frac{1}{\sqrt{1-x^2}}$ d) $\frac{1}{1-x^2}$
8. In greater than or equal to constraints, the non-negative variable that is used to balance both side is
a) Condition variable b) Surplus variable
c) Slack variable d) Solving variable

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1. c. $2u$

2. d. consistent and has infinitely solution

3. c. $n+2$

4. d. -8

5. b. $-\frac{1}{3}$

6. b. 5

7. d. $\frac{1}{1-x^2}$

8. c. slack variable

9. a. 3cm/s

10. a. $\frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$

11. c. $\frac{\pi}{4}$

9. The volume of a sphere is increasing at the rate of $3\pi \text{ cm}^3/\text{sec}$. The rate of change of its radius when radius $\frac{1}{2} \text{ cm}$ is

- a) 3 cm/s b) 2 cm/s c) 1 cm/s d) $\frac{1}{2} \text{ cm/s}$

10. $\int \frac{dx}{x^2 - a^2} = ?$

a) $\frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + c$

c) $\frac{1}{a} \ln \left| \frac{x-a}{x+a} \right| + c$

b) $\frac{1}{2a} \ln \left| \frac{x+a}{x-a} \right| + c$

d) none

11. If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = \sqrt{6}$, then the angle between $\vec{a}$ and $\vec{b}$ is

a) $\frac{\pi}{6}$

b) $\frac{\pi}{3}$

c) $\frac{\pi}{4}$

d) $\frac{2\pi}{6}$

GROUP - B [8×5=40]

12. Define Permutation and combinations. A committee of 5 persons is to be selected from 5 men and 4 ladies. In how many ways can this be done so that at least two ladies are always included? [2+3]

13. Define vector product of two vectors $\vec{a}$ and $\vec{b}$. Interpret it geometrically. If $\vec{a} = \vec{i} + 3\vec{j} + 2\vec{k}$ and $\vec{b} = 2\vec{i} + 3\vec{j} + \vec{k}$ then find the area of parallelogram whose adjacent sides represented by $\vec{a}$ and $\vec{b}$. [1+2+2]

14. State De-moivre's theorem. By using it, find cube roots of unity and show that the sum of cube roots of unity is zero. [1+3+1]

15. a) Find $\frac{dy}{dx}$, if $y = (\sinh x)^{\cos h^{-1} x}$ [3]

b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$ [2]

16. a) Evaluate: $\int \frac{dx}{1-2\cos x}$

b) Evaluate: $\int \frac{dx}{(2x+1)\sqrt{4x+5}}$

[2+3]

17. a) Is the System $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ well-conditioned? Justify your answer. [2]

b) By using vector method, Prove that: $\sin(A+B) = \sin A \cos B + \cos A \sin B$ [3]

18. Solve the following LP Problem, using Simplex method
Maximize $Z = 3x + 4y$

subject to constraints

$$2x + y \leq 8, \quad x + 3y \leq 15, \quad x, y \geq 0$$

[1+4]

19. a) Using the Gauss elimination method, solve the following system of equation. [3]

$$x - 2y + 3z = 2, \quad 2x - 3y + z = 1, \quad 3x - y + 2z = 9$$

b) If θ be angle between $\vec{a}$ and $\vec{b}$, then prove that:

$$(\vec{a} \times \vec{b})^2 + (\vec{a} \cdot \vec{b})^2 = a^2 b^2$$

[2]

GROUP-C [3×8=24]

20. a) Write the expression for $\sinh x$ and $\cosh x$ in terms of exponential function. [1]

b) Define L Hospital's rule. By using it, evaluate: $\lim_{x \rightarrow 0^+} (x^2 \ln x)$ [1+2]

c) The radius of a spherical balloon is increasing at the rate of 10 cm/sec. At what rate is the surface area of the balloon increasing when the radius is 15 cm? [2]

d) Find the points on the curve $y = x^3 - 3x^2 - 9x + 7$ where the tangents are parallel to the x-axis. [2]

21. a) Let $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ are two linear equations.

i) Write the expression in matrix form.

ii) Under which conditions the above system of linear equations may have infinite solution, Unique solution, and no solution? [2]

b) Solve the equation by using inverse matrix method or row equivalent method

$$3x - y - 2z = 1, \quad x - y + 2z = 3, \quad -2x + 3y + z = 8$$

[3]

c) Find the number of arrangements of the letters of the word "LAPTOP" so that:

i) They always begin with L and end with T.

ii) They do not begin with L but always end with T? [3]

22. a) Solve the following system of linear Equation by using Gauss-Seidel method.

[4]

✓ $5x_1 - 2x_2 + 3x_3 = -1,$

$-3x_1 + 9x_2 + x_3 = 2,$

$2x_1 - x_2 - 7x_3 = 3$

b) In the expansion of $\left(x^2 - \frac{6}{x}\right)^8$, Find

✓ i) The number of terms.

ii) The middle term.

✓ iii) The general term

iv) The coefficient of x^4 (if any)

[4]

Best of luck!

KMC Note